ABSTRACT

AI-Based Hotel Kitchen Cleanliness Monitoring System

Maintaining cleanliness and hygiene in hotel kitchens is essential to ensure food safety, protect public health, and uphold the reputation of hospitality establishments. Traditional kitchen inspections are typically manual, time-consuming, and subjective, often resulting in inconsistent hygiene evaluations. To address these limitations, this project presents an AI-based Hotel Kitchen Cleanliness Monitoring System that automatically evaluates kitchen hygiene using deep learning and image processing techniques.

The proposed system uses labeled image datasets of hotel kitchens categorized as *clean* and *unclean*. The images undergo preprocessing and are analyzed using the YOLOv8 deep learning architecture, which enables accurate and real-time image recognition and classification.

The backend of the system is developed using Python with the Django framework, allowing users to upload kitchen images and receive instant cleanliness predictions. The results are stored in a SQLite database, ensuring efficient data storage and retrieval. This vision-based approach reduces dependence on human inspectors and physical sensors, providing an automated, unbiased, and scalable solution for monitoring kitchen hygiene in real time.

Dataset

- Name**:** Kitchen Hygiene Dataset
- Source Link**:** https://universe.roboflow.com/kitchen-hygiene-efuu5/kitchenhygiene

Technology Stack

- Frontend: HTML, CSS, Bootstrap
- Backend: Python, Django
- Deep Learning & Computer Vision**:** YOLOv8, PyTorch, OpenCV, NumPy
- Database: SQLite (DB Browser for SQLite)
- IDE & Tools: VS Code, Google Colab, Roboflow

References

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